# Contents

1

# Smooth Manifold

## 1.1 Topological Manifolds

**Definition 1.1.1**

Suppose $M$ is a topological space. We say that $M$ is a ***topological manifold of dimension n*** or a ***topological n-manifold*** if it has the following properties:

- $M$ is a ***Hausdorff space***$(\mathrm{T}_2)$ : for every pair of distinct points $p, q \in M$, there are disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- $M$ is ***second-countable***: there exists a countable basis for the topology of $M$.
- $M$ is ***locally Euclidean of dimension n***: each point of $M$ has a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

*Remark.*
The third property means, more specifically, that for each $p \in M$ we can find

- an open subset $U \subseteq M$ containing $p$,
- an open subset $\hat{U} \subseteq \mathbb{R}^n$, and
- a homeomorphism $\varphi : U \to \hat{U}$.

*Remark.*
A topological manifold $M$ is locally compact, locally path-connected, seperable, Lindëlof, $\sigma$-finite, $T_3$, $T_4$, and paracompact.

### 1.1.1 Coordinate Chart

**Definition 1.1.2 ▸ Coordinate Chart**

Let $M$ be a topological ***n-manifold***. A ***coordinate chart*** (or just a ***chart***) on $M$ is a pair $(U, \varphi)$, where $U$ is an open subset of $M$ and $\varphi : U \to \hat{U}$ is a homeomorphism from $U$ to an open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$ .

*Remark.*

- By definition of a topological manifold, each point $p \in M$ is contained in the domain of some chart $(U, \varphi)$.
- If $\varphi(p) = 0$, we say that the chart is ***centered at p***.
- If $(U, \varphi)$ is any chart whose domain contains $p$, it is easy to obtain a new chart centered at $p$ by subtracting the constant vector $\varphi(p)$.

## 1.2 Smooth Structures

**Definition 1.2.1 ▸ Smoothly Compatible**

Let $M$ be a topological $n$-manifold.

- If $(U, \varphi), (V, \psi)$ are two charts such that $U \cap V \neq \varnothing$, the composite map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is called the ***transition map from $\varphi$ to $\psi$***.
- Two charts $(U, \varphi)$ and $(V, \psi)$ are said to be ***smoothly compatible*** if either $U \cap V = \varnothing$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.

*Remark.*

- $\psi \circ \varphi^{-1}$ is a composition of homeomorphisms, and is therefore itself a homeomorphism.
- Since $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open subsets of $\mathbb{R}^n$, smoothness of this map is to be interpreted in the ordinary sense of having continuous partial derivatives of all orders.

**Definition 1.2.2 ▸ Atlas**

We define an ***atlas for M*** to be a collection of charts whose domains cover $M$. An atlas $\mathcal{A}$ is called a ***smooth atlas*** if any two charts in $\mathcal{A}$ are smoothly compatible with each other.

**Definition 1.2.3 ▸ Maximal Atlas**

A smooth atlas $\mathcal{A}$ on $M$ is ***maximal*** if it is not properly contained in any larger smooth atlas. This just means that any chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

**Definition 1.2.4 ▸ Smooth Structures and Smooth Manifolds**

- If $M$ is a topological manifold, a ***smooth structure on M*** is a maximal smooth atlas.
- A ***smooth manifold*** is a pair $(M, \mathcal{A})$, where $M$ is a topological manifold and $\mathcal{A}$ is a smooth structure on $M$.

**Lemma 1.2.5 ▸ Smooth Manifold Chart Lemma**

Let $M$ be a set, and suppose we are given a collection $\{U_\alpha\}$ of subsets of $M$ together with maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$, such that the following properties are satisfied:

1. For each $\alpha$, $\varphi_\alpha$ is a bijection between $U_\alpha$ and an open subset $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.
2. For each $\alpha$ and $\beta$, the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$.
3. Whenever $U_\alpha \cap U_\beta \neq \emptyset$, the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth.
4. Countably many of the sets $U_\alpha$ cover $M$.
5. Whenever $p, q$ are distinct points in $M$, either there exists some $U_\alpha$ containing both $p$ and $q$ or there exist disjoint sets $U_\alpha, U_\beta$ with $p \in U_\alpha$ and $q \in U_\beta$.

Then $M$ has a unique smooth manifold structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

## 1.3 Manifolds with Boundary

**Definition 1.3.1 ▸ Manifolds with Boundary**

An ***n-dimensional topological manifold with boundary*** is a second-countable Hausdorff space $M$ in which every point has a neighborhood homeomorphic either to an open subset of $\mathbb{R}^n$ or to a (relatively) open subset of $\mathbb{H}^n$ .

**Definition 1.3.2 ▶ Chart for Manifolds with Boundary**

Let $M$ be an n-dimensional topological manifold with boundary. An open subset $U \subseteq M$ together with a map $\varphi : U \to \mathbb{R}^n$ that is a homeomorphism onto an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ will be called a ***chart for*** $M$.

- When it is necessary to make the distinction, we will call $(U, \varphi)$ an ***interior chart*** if $\varphi(U)$ is an open subset of $\mathbb{R}^n$ (which includes the case of an open subset of $\mathbb{H}^n$ that does not intersect $\partial\mathbb{H}^n$).
- We call $(U, \varphi)$ a ***boundary chart*** if $\varphi(U)$ is an open subset of $\mathbb{H}^n$ such that $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$.
- A boundary chart whose image is a set of the form $B_r(x) \cap \mathbb{H}^n$ for some $x \in \partial\mathbb{H}^n$ and $r > 0$ is called a ***coordinate half-ball***.